

Helixo AI: An Intelligent Edge-Computing Framework for Real-Time Helmet Compliance and Proactive Ignition Interlock Control

PROBLEM STATEMENT

Road traffic injuries remain a leading cause of mortality globally, with two-wheeler riders disproportionately affected, particularly in developing nations like India. The vast majority of these fatalities are attributed directly to severe head trauma resulting from the failure to wear safety helmets. Despite stringent legislative mandates and associated financial penalties, voluntary compliance among riders remains critically low.

The fundamental flaw in contemporary traffic enforcement paradigms—such as manual policing or automated Traffic Enforcement Cameras (TECs)—is their inherently reactive nature. These systems issue citations post-facto, penalizing the violation only after it has occurred, while doing absolutely nothing to physically mitigate the immediate, life-threatening risk posed by unhelmeted riding. Furthermore, existing technological propositions, such as RFID or Bluetooth-embedded smart helmets, require specialized hardware modifications to the helmet itself, rendering them incompatible with standard, off-the-shelf equipment and impractical for mass adoption.

Therefore, there exists a critical need for a proactive, intelligent, and cost-effective vehicular safety framework that operates independently of the helmet type. The system must enforce compliance at the source by physically restricting vehicle operation—specifically, the ignition interlock—unless a helmet is actively detected on the rider. This requires a robust, offline, edge-computing solution capable of real-time computer vision inference directly on the

motorcycle, immune to the latency and connectivity issues associated with cloud-based architectures.